

CSG Sweeps: Feature-aware sweep surfacing of deformable CSGs (Supplemental Materials)

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1 Introduction

In this document, we provide proofs of the theorems in the paper (Section 2) and additional technical details, including the analysis of sharp features on the sweep surface (Section 3), arrangement of oriented cell complexes (Section 4), and soft booleans for comparing with [Ju et al. 2025] (Section 5).

2 Theorem proofs

2.1 Proof of Theorem 4.1

The correctness of the statement in the case of a C^2 -continuous sweep function f is already proved in [Ju et al. 2025] (Theorem 4.1), using their definition of the outward orientation of the lifted envelope. To prove the statement on a C^0 -continuous CSG sweep function f , our idea is to perturb f to obtain a smooth function $\tilde{f}$, and to show that our definition of the outward orientation for the lifted envelope of f agrees with the outward orientation defined in [Ju et al. 2025] for the lifted envelope of $\tilde{f}$, as $\tilde{f}$ approaches f .

PROOF. A CSG sweep function f in $\mathbb{R}^{n+1}$ is C^2 -continuous away from some subspace Q dominated by two or more member functions, where f is only C^0 . By genericity, Q has dimension n . Consider a C^2 -continuous function $\tilde{f}$ that differs from f only within an ϵ -neighborhood of Q , in which $\tilde{f}$ and its partial derivatives (up to 2nd order) differ from those of f by at most ϵ (away from Q). By this construction, the lifted envelope of $\tilde{f}$, as well as its tangent spaces, converge to those of f as $\epsilon \rightarrow 0$.

Our goal is to show that, as $\epsilon \rightarrow 0$, an outward tangent basis on the lifted envelope of $\tilde{f}$, as defined in [Ju et al. 2025], remains

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outward on the lifted envelope of f according to our definition in Section 4.3. Since the two lifted envelopes differ only within the ϵ -neighborhood of the sharp contours of f , and since our definition of an outward orientation agrees with [Ju et al. 2025] on other parts of the lifted envelope (smooth contours and caps), we focus our attention on the sharp contour of f .

Consider a point p on the sharp contour of f dominated by functions f_1, f_2 , such that (without loss of generality) $f'_1(p) > 0 > f'_2(p)$. Let $\tilde{p}$ be the closed point of p on the contour part of the lifted envelope of $\tilde{f}$ (i.e., p is not on the caps), and $\tilde{\Sigma}$ an outward tangent basis of the lifted envelope at $\tilde{p}$. By [Ju et al. 2025],

$$\det(\{\tilde{\Sigma}, \nabla \tilde{f}(\tilde{p}), \nabla \tilde{g}(\tilde{p})\}) > 0, \quad (1)$$

where $\tilde{g}(\tilde{p}) = \tilde{f}(\tilde{p})$ (since $\tilde{p}$ is not on the caps).

We will examine the two gradients, $\nabla \tilde{f}(\tilde{p})$ and $\nabla \tilde{g}(\tilde{p})$, as $\tilde{f}$ approaches f . Since $\tilde{f}$ is smooth, both gradients are normal to the level sets of the respective functions, $\tilde{f}$ and $\tilde{g}$. Whereas the level set of f has a sharp corner at p , as the surface normal jumps from $\nabla f_1(p)$ to $\nabla f_2(p)$, the level set of $\tilde{f}$ forms a rounded corner at $\tilde{p}$. As $\epsilon \rightarrow 0$, $\nabla \tilde{f}(\tilde{p})$ lies in the “cone” between $\nabla f_1(p)$ to $\nabla f_2(p)$ and can be expressed as their convex combination,

$$\nabla \tilde{f}(\tilde{p}) \rightarrow \mu \nabla f_1(p) + \lambda \nabla f_2(p), \quad (2)$$

for some positive scalars μ, λ .

The level set of $\tilde{g}$ approaches the level set of g (the silhouette set) as $\epsilon \rightarrow 0$. As discussed in Section 4.2, the part of the silhouette set containing the sharp contour point p is the sharp contour set, which lies on the zero-level set of the difference $f_1 - f_2$. This level set, denoted by S , has the normal direction $v = \nabla f_1(p) - \nabla f_2(p)$ at p . As $\epsilon \rightarrow 0$, $\nabla \tilde{g}(\tilde{p})$ becomes parallel to v and points to the side of S where g is positive. Observe that,

$$v \cdot \vec{t} = \nabla f_1(p) \cdot \vec{t} - \nabla f_2(p) \cdot \vec{t} = f'_1(p) - f'_2(p) > 0,$$

where $\vec{t}$ is the time direction. This implies that v points to the side “above” S (in time). This side is positive (resp. negative) in g if f_1 (resp. f_2) is dominating on that side. On the other hand, $f'_1(p) > f'_2(p)$ and $f_1(p) = f_2(p)$ imply that f_1 is greater (resp. smaller) than f_2 above (resp. below) p . Therefore, f_1 (resp. f_2) is dominating on the above side of S if the operator $\square$ at the root (as defined in Section 4.3) of f_1, f_2 in the CSG tree is $\max$ (resp. $\min$). We conclude that, as $\epsilon \rightarrow 0$, $\nabla \tilde{g}(\tilde{p})$ approaches the direction v if $\square$ is $\max$ and $-v$ otherwise. That is,

$$\nabla \tilde{g}(\tilde{p}) \rightarrow \begin{cases} \gamma(\nabla f_1(p) - \nabla f_2(p)), & \text{if } \square = \max \\ \gamma(\nabla f_2(p) - \nabla f_1(p)), & \text{if } \square = \min \end{cases} \quad (3)$$

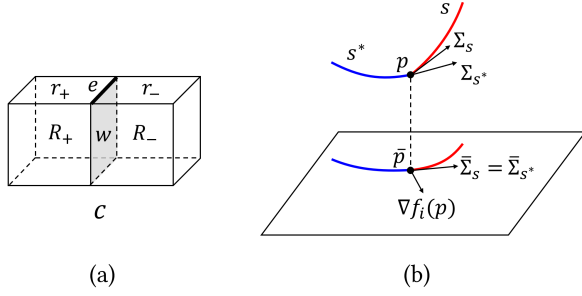


Fig. 1. Illustrations for the proof of Lemma 2.1 (a) and sharp feature analysis (b).

for some positive scalar γ .

Finally, substituting Equations 2 and 3 into 1 yields, as $\epsilon \rightarrow 0$,

$$\begin{cases} \det(\{\tilde{\Sigma}, \nabla f_1(p), \nabla f_2(p)\}) < 0, & \text{if } \square = \max \\ \det(\{\tilde{\Sigma}, \nabla f_1(p), \nabla f_2(p)\}) > 0, & \text{if } \square = \min \end{cases} \quad (4)$$

By our orientation definition in Section 4.3, $\tilde{\Sigma}$ is an outward orientation of the lifted envelope of f at p . $\square$

The proof shows that our orientation rule on sharp contours is the same as the orientation rule in [Ju et al. 2025] on a slightly perturbed, *smooth* contour. As the perturbation does not change the global structure of the lifted envelope or silhouette set, the following alternative orientation rule for a smooth sweep from [Ju et al. 2025] also works for a CSG sweep: the outward-oriented lifted envelope faces the subspace of the silhouette set where the sweep function f is positive, if the silhouette set is oriented to face the subspace of $\mathbb{R}^{n+1}$ where the silhouette function g is positive.

2.2 Proof of Theorem 6.1

We start with an observation of the cell complex C' produced by our algorithm that cuts an n -dimensional cell complex C by the zero-level set of a function h .

LEMMA 2.1. *Each k -wall in C' , for $k \in [0, n-1]$, is a face of exactly two $(k+1)$ -rooms with positive and negative h signs.*

PROOF. We prove by induction on k . The statement holds trivially in the base case of $k = 0$, as a 0-wall always divides a 1-cell in C into two 1-rooms with different h signs (Figure 9 (a)).

Suppose the statement holds for any $k < l$ where $l \leq n$, we will show that it holds for $k = l$. Consider an l -wall w that splits an $(l+1)$ -cell $c \in C$ (see Figure 1 (a)). By our algorithm, a face e of w is an $(l-1)$ -wall of a face f of c . By the induction hypothesis, e is a face of two l -rooms in f , denoted by r_+ , r_- , that are respectively positive and negative in h . Due to their opposite signs and being connected with w , r_+ and r_- are faces of two $(l+1)$ -rooms in c , denoted by R_+ , R_- , that are respectively positive and negative in h and both have w as a face. Consider any other face e' of w , which lies on a face f' of c and is a face of l -rooms r'_+ , r'_- in f' . As r'_+ (resp. r'_-) is connected to w , it is also a face of R_+ (resp. R_-). As a result, w is a face of exactly two $(l+1)$ -rooms, R_+ , R_- , with opposite h signs. $\square$

We next show that cutting a manifold subset of C maintains its manifoldness and that the cut is a manifold of a lower dimension. A set of k -cells (with their boundary cells) S is a k -manifold if each $(k-1)$ -cell in S is a face of one or two k -cells in S . The *boundary* of S , denoted by ∂S , consists of all $(k-1)$ -cells of S having only one incident k -cell. A k -manifold S is *closed* if ∂S is empty.

LEMMA 2.2. *Let S be a k -manifold in $\mathbb{R}^n$, for some $k \leq n$, and S' be the cell complex cut by the zero-level set of some function h , then:*

- (1) S' is a k -manifold whose boundary coincides with ∂S .
- (2) The union of zero-signed cells in S' is a $(k-1)$ -manifold whose boundary lies on ∂S .

PROOF. We will prove each statement.

- (1) Consider a $(k-1)$ -cell $c \in S'$. There are three cases:

- (a) c is an original cell in S : As S is a k -manifold, c is a face of one or two k -cells of S , and the former happens if $c \in \partial S$. By our algorithm, each of these one or two k -cells is either unsplit or split into multiple k -rooms, one of which contains c . As a result, c is a face of one or two k -cells in S' , with the former happening if $c \in \partial S$.
- (b) c is a $(k-1)$ -room in some $(k-1)$ -cell $c^* \in S$: As S is a k -manifold, c^* is a face of one or two k -cells of S , each split into multiple k -rooms among which one contains c . So c is a face of one or two k -cells in S' , with the former happening if c^* (and hence c) is on ∂S .
- (c) c is a $(k-1)$ -wall in some k -cell $c^* \in S$: By Lemma 2.1, c is a face of two k -rooms in c^* .

In sum, c is a face of two k -cells in S' unless c is on ∂S , in which case c is a face of one k -cell in S' .

- (2) Let L be the union of zero-signed cells in S' . By our algorithm, L consists of walls at dimensions $k-1$ and lower. Consider a $(k-2)$ -cell $w \in L$, which is a $(k-2)$ -wall in some $(k-1)$ -cell $c \in S$. As S is a k -manifold, c is a face of one or two k -cells of S , and the former happens if $c \in \partial S$. By our algorithm, w is a face of exactly one $(k-1)$ -wall in each of these one or two k -cells, and these walls are the only $(k-1)$ -cells in L incident on w . As a result, w is a face of one or two $(k-1)$ -cells in L , with the former happening if c (and hence w) is on ∂S . $\square$

We are now ready to prove Theorem 6.1.

PROOF. We prove by induction on k , the number of functions in H whose level sets intersect. In the base case of $k = 1$, H consists of a single function h_i . As C is an n -manifold, Lemma 2.2(1) ensures that each cutting produces an n -manifold with the same boundary as ∂C . Consider the cell complex after processing h_i , which we denote by C_i . By Lemma 2.2(2), the zero-level set of h_i in C_i , denoted by L , is an $(n-1)$ -manifold whose boundary is on ∂C_i (and hence on ∂C). On the other hand, Lemma 2.2(1) ensures that each subsequent cutting preserves the manifoldness of L as well as its boundary. As a result, the zero-level set of h_i after all cutting is complete remains an $(n-1)$ -manifold whose boundary lies on ∂C .

Suppose the statement holds for all $k < l$ for some $l \leq \min(n, m)$, we will show that it holds for $k = l$. Let h_l be the last function in H that is processed for arrangement, and consider the cell complex C_l

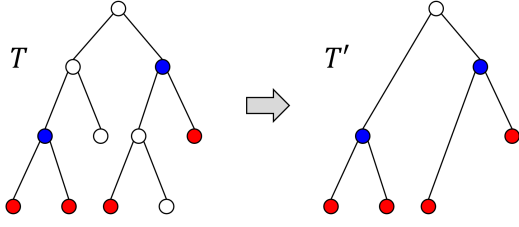


Fig. 2. Pruning a binary tree T for a given set of leaf nodes V_1 (red) and internal nodes V_2 (blue).

after processing h_i . By the induction hypothesis, the intersection L' of the zero-level sets of $l - 1$ functions $H' = H \setminus \{h_i\}$ in C_{i-1} is an $(n - l + 1)$ -manifold whose boundary lies on ∂C . The intersection of the zero-level sets of all functions in H is the part of the zero-level set of h_i in C_i that lies on L' , which we denote as L . Note that L is the same as the result of cutting L' by the zero-level set of h_i . By Lemma 2.2, L is an $(n - l)$ -manifold whose boundary is on $\partial L'$ and hence on ∂C . Finally, Lemma 2.2(1) ensures that each subsequent cutting preserves the manifoldness of L as well as its boundary, and hence the intersection of all zero-level sets of H after computing the arrangement is an $(n - l)$ -manifold whose boundary lies on ∂C . $\square$

2.3 Proof of Theorem 6.2

We start with an observation of a binary tree. We define the *root* of two leaf nodes v_1, v_2 in a binary tree T as the unique internal node v_3 of T where v_1, v_2 reside in the subtrees of different children of v_3 .

LEMMA 2.3. *Consider a set V_1 of leaf nodes and a set V_2 of internal nodes of a binary tree T . If every node in V_2 is the root of some pair of nodes in V_1 , then $|V_1| > |V_2|$.*

PROOF. We construct a “pruned” tree T' from T as follows. The leaves of T' are V_1 . The internal nodes of T' are the set of all internal nodes in T such that each is the root of some pair of nodes in V_1 . Two nodes in T' are edge-connected if they are connected by a path in T that does not go through any other nodes in V_2 . See an example in Figure 2.

Note that T' is also a binary tree. This is because each internal node v of T' is the root of two leaf nodes in T' , meaning that v must have both left and right children. As a result, the number of leaf nodes of T' (i.e., $|V_1|$) is greater than the number of internal nodes. On the other hand, V_2 is a subset of T' 's internal nodes. It follows that $|V_1| > |V_2|$. $\square$

Next, we prove Theorem 6.2.

PROOF. We prove each statement in turn. Recall that the sweep function f is defined by member functions $f_1, \dots, f_m$ in $\mathbb{R}^{n+1}$ and a binary CSG tree T , whose internal nodes are $v_1, \dots, v_{m-1}$. Let r_i be the node function at an internal node v_i , and S_i be the equal set of the (member or node) functions at the two child nodes of v_i .

- (1) A point x lies on the equal set S_i for some $i \in [1, m - 1]$ if there are two member functions (leaves of T) whose root is v_i , and that both functions are dominant at x in r_i . Note that any pair of member functions that are dominant in some node

function must be dominant in the node function at their root node. On the other hand, a point x on the sharp contour set has (at least) two dominant functions in the sweep function f , which is the node function at the root of T . Let v_i be the root node of those two dominant functions; then x lies on S_i .

- (2) A point x lies on the equal set S_i is a solution to the equation $f_p(x) = f_q(x)$ where f_p, f_q are two member functions whose root is v_i . Points that lie on the intersection of k equal sets $\{S_i\}_{i \in I}$ are solutions to k equations. To show that such a solution space forms an $(n - k + 1)$ -manifold, and assuming $f_1, \dots, f_m$ satisfy the genericity assumptions in Section 4.1, it suffices to show that any subset of $l \leq k$ equations involves at least $l + 1$ member functions. Let V_2 be the set of internal nodes corresponding to the l equations, and V_1 be the leaf nodes representing the member functions involved in these equations. As such, each node in V_2 is the root of some pair of nodes in V_1 . By Lemma 2.3, the size of V_1 is at least $l + 1$, concluding the proof. $\square$

3 Sharp features on the sweep boundary

The sweep boundary in $\mathbb{R}^n$ is a piecewise-smooth $(n - 1)$ -manifold, where smooth patches meet with tangential discontinuity at *sharp features*. These sharp features, in turn, are made up of smooth manifolds at lower dimensions. We will analyze and classify $(n - 2)$ -dimensional sharp features on a sweep boundary, which we call *edges* (e.g., sharp curves on a smooth 2-manifold). These edges, in turn, meet at sharp features at lower dimensions. We leave the analysis of *isolated* lower-dimensional sharp features, which are not incident on an edge, as future work. All our discussions assume the input is generic (i.e., in general position).

We focus on sharp edges on the envelope away from its self-intersections. To this end, consider a *segment* on the lifted envelope in $\mathbb{R}^{n+1}$ as a connected component of the same type (smooth contour, sharp contour, bottom cap, or top cap), such that all points on the segment have the same dominant function (for the smooth contour and caps) or the same pair of dominant functions (for the sharp contour). Note that each segment is the intersection of two smooth n -manifolds: a segment of the smooth contour, bottom cap, or top cap with dominant function f_i lies on the intersection of the zero-level set of f_i with the zero-level set of f'_i , the bottom time plane ($t = 0$), or the top time plane ($t = 1$), while a segment of the sharp contour with dominant functions $\{f_i, f_j\}$ lies on the intersection of the zero-level sets of these two functions. As a result, each segment is a smooth $(n - 1)$ -manifold in $\mathbb{R}^{n+1}$.

Next, we show that the projection of each segment to $\mathbb{R}^n$ does not produce sharp edges on the sweep boundary. This is obviously true for segments on the bottom or top cap, which are identical to their projection. The projection of smooth or sharp contours may be non-smooth at *singularities*, where the tangent space on the contour “collapses” (i.e., spans less than $n - 1$ dimensions) after projection. For smooth contours, [Ju et al. 2025] showed that singular points on the envelope that form an $(n - 2)$ -manifold (called a *cusp*) lie inside the sweep volume and hence do not appear on the sweep boundary. Now consider a sharp contour segment dominated by functions $\{f_i, f_j\}$.

The tangent basis Σ at a point p on the segment is orthogonal to both ∇f_i and ∇f_j . On the other hand, p projects to a singular point when the time direction $\vec{t}$ lies in the space spanned by Σ . This implies that both ∇f_i and ∇f_j are orthogonal to $\vec{t}$. We therefore have equalities $f_i(p) = f_j(p) = 0$ and $\nabla f_i(p) \cdot \vec{t} = \nabla f_j(p) \cdot \vec{t} = 0$. Due to genericity, the solution to these 4 equations in $\mathbb{R}^{n+1}$ spans a subspace with no more than $n - 3$ dimensions, so its projection on the envelope (the singular points) has less than $n - 2$ dimensions.

Finally, we show that the boundaries of smooth contour segments do not project to sharp edges. Consider a point p on a smooth contour segment s dominated by f_i . The tangent basis Σ_s of s at p is orthogonal to $\nabla f_i(p)$, which in turn is orthogonal to the time direction (by the definition of the smooth contour). Assuming the projection of p on the envelope, $\bar{p}$, is not singular, the tangent basis of the envelope at $\bar{p}$, denoted by $\bar{\Sigma}_s$, is also orthogonal to $\nabla f_i(p)$ (see Figure 1 (b)). Suppose p is on the boundary between s and another segment s^* . Due to genericity, s^* is either a cap segment dominated by f_i or a sharp contour segment dominated by $\{f_i, f_j\}$ for some $j \neq i$. In either case, the tangent basis Σ_{s^*} of s^* at p is also orthogonal to $\nabla f_i(p)$, which (at the location p) is orthogonal to the time direction. As a result, the projected tangent basis, $\bar{\Sigma}_{s^*}$, is also orthogonal to $\nabla f_i(p)$ (see insert). As the projections of segments s, s^* share the same $(n - 1)$ -dimensional tangent space at $\bar{p}$ (both orthogonal to $\nabla f_i(p)$), the two projected segments are G^1 continuous at $\bar{p}$.

We conclude that the sharp edges on the sweep boundary (away from the envelope self-intersections) are projections of the boundaries between cap or sharp contour segments. As the cap segments lie at the beginning and end of the sweep, boundaries between two cap segments or between a cap segment and a sharp contour segment project to the original sharp edges on the shape (the *sweep-end* features). On the other hand, boundaries between two sharp contour segments project to sharp edges created during sweeping (the *in-sweep* features).

4 Oriented cell complex and arrangement

We will augment the cell complex representation with orientations, and describe how to maintain consistent orientations in our arrangement algorithm described in Section 6.1.

The orientation of a cell c is expressed by a binary-valued *orientation function* $o(f, c)$ for each face f of c . A value of $+1$ indicates that f should maintain its original orientation, while -1 means f should be “flipped”. Note that different cells sharing a common face may assign different orientations to that face. Intuitively, the orientation of c is consistent if all its faces, after flipping, are oriented towards either inside or outside c . In addition, for two adjacent n -cells in $\mathbb{R}^n$, their shared faces should be assigned opposite orientations.

To formalize the notion of consistency, we first define a *valid cell complex* in $\mathbb{R}^n$ as one that meets the following criteria (V1-3):

- V1:** Each 1-cell has two faces (vertices).
- V2:** Faces of each k -cell ($k \in [2, n]$) form a closed k -manifold.
- V3:** The cell complex is an n -manifold.

The orientation function o is *consistent* on a valid cell complex C if the following criteria are met (C1-3). We call a face of a face of a k -cell c , which is a $(k - 2)$ -cell, a *subface* of c .

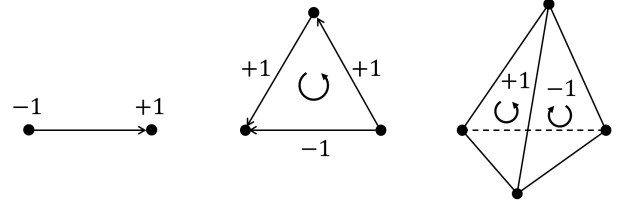


Fig. 3. Consistent orientations of a 1-cell, 2-cell, and 3-cell. Orientation function values are noted on each face of the cell (only front-facing faces of the 3-cell). Each edge vector is drawn from the -1 end to the $+1$ end. The winding on a triangle shows the edge directions after flipping.

C1: For each 1-cell c with faces f_1, f_2 ,

$$o(f_1, c) = -o(f_2, c).$$

C2: For each k -cell ($k \in [2, n]$) c and each pair of its faces f_1, f_2 sharing a subface e ,

$$o(e, f_1) * o(f_1, c) = -o(e, f_2) * o(f_2, c).$$

C3: For each pair of n -cells c_1, c_2 sharing a common face f ,

$$o(f, c_1) = -o(f, c_2).$$

Figure 3 shows examples of consistently oriented cells.

We next discuss how to maintain a consistent orientation during the arrangement computation. First, we show that cutting maintains the validity of the cell complex.

LEMMA 4.1. *The cell complex C' resulting from cutting a valid cell complex C is also valid.*

PROOF. The validity criterion **V3** (an n -manifold) is met by C' due to the manifold-ness of C and Lemma 2.2(1). We will show that C' meets the other two criteria by examining cells at different dimensions. We only consider the new cells – rooms and walls – introduced by cutting. Cells of C that remain after cutting, by definition, contain only original boundary cells from C , and therefore meet the requirements due to the validity of C .

To verify criterion **V1**, note that a 1-cell $c \in C'$ is either a 1-room, which is split from a 1-cell in C , or a 1-wall, which connects two 0-walls on the edges of a 2-cell in C . In either case, c has two vertices, and the requirement is met.

To verify criterion **V2**, we need to show that, for each k -cell ($k \in [2, n]$) $c \in C'$, each subface of c is a face of exactly two faces of c . We will consider the two types of cell c :

- (1) c is a k -wall in a $(k + 1)$ -cell $c^* \in C$: Each subface of c is a $(k - 2)$ -wall on a subface of c^* . Consider a subface e of c that lies on a subface e^* of c^* (Figure 4 (a)). As C is valid, e^* is a face of two faces of c^* , denoted by f_1^* and f_2^* . Each of them contains a $(k - 1)$ -wall, denoted by f_1 and f_2 , in which e is a face. As f_1, f_2 are also faces of c , e is a face of two faces of c .
- (2) c is a k -room in a k -cell $c^* \in C$: Each subface e of c is either in a subface of c^* (the subface itself or a room of it) or a $(k - 2)$ -wall in a face of c^* . In the former case, suppose e is in a subface e^* of c^* (Figure 4 (b)). Due to the validity of C , e^* is a face of two faces of c^* , denoted by f_1^*, f_2^* , each of which is either unsplit or contains a $(k - 1)$ -room with e as its face.

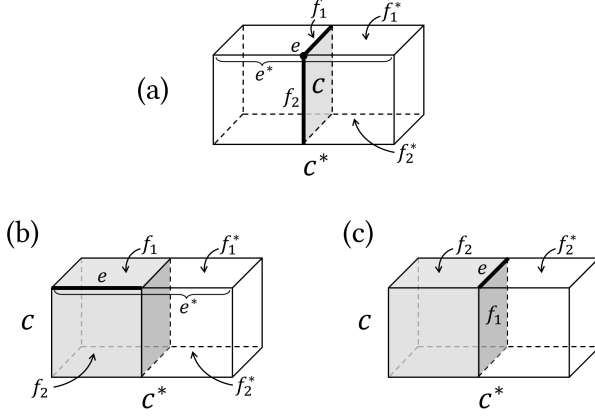


Fig. 4. Illustrations for Lemmas 4.1 and 4.2.

So e is a face of two $(k-1)$ -rooms that are faces of c . In the latter case, let f_2^* be a face of c^* containing e (Figure 4 (c)). By Lemma 2.1, e is a face of exactly one $(k-1)$ -room of f_2^* with the same sign as c , which we denote by f_2 . As a result, f_2 is a face of c . On the other hand, e is also a face of a $(k-1)$ -wall of c^* , f_1 , which is another face of c . As a result, e is a face of two faces of c , f_1 and f_2 .

□

Given a valid cell complex C with a consistent orientation o , we will define a new orientation function o' on the complex C' resulting from cutting C by the zero-level set of some function h . First of all, if a cell $c \in C'$ is already in C (i.e., c is unsplit), then each face f of c is also in C , and we simply inherit the original orientation by setting $o'(f, c) = o(f, c)$. We assign orientations in a newly created k -room ($k \in [1, n]$) or k -wall ($k \in [1, n-1]$) $c \in C'$ as follows:

- c is a k -room in a k -cell $c^* \in C$:
 - $k = 1$: Let f_1, f_2 be the vertices of c . If f_1 is a face of c^* , it inherits its orientation from c^* , whereas f_2 (the 0-wall) gets the opposite orientation (Figure 5 (a)). That is,

$$o'(f_1, c) = o(f_1, c), \quad o'(f_2, c) = -o'(f_1, c).$$

- $k > 1$: A face f of c is either a wall in c^* or lies in a face f^* of c^* . In the former case, we orient f towards (resp. away from) c if the h sign of c is positive (resp. negative). That is,

$$o'(f, c) = \text{sign}(h(c)).$$

In the latter case, f inherits the orientation of f^* in c^* :

$$o'(f, c) = o(f^*, c^*).$$

- c is a k -wall in a $(k+1)$ -cell $c^* \in C$:
 - $k = 1$: Let f_1, f_2 be the vertices of c . They are 0-walls on edges f_1^*, f_2^* of c^* . Denote by r_i ($i = 1, 2$) the 1-room in f_i^* with positive h (Figure 5 (b)). For $i = 1, 2$, we set

$$o'(f_i, c) = -o(f_i^*, c^*) * o'(f_i, r_i).$$

- $k > 1$: A face f of c is a wall in a face f^* of c^* . We obtain the orientation of f in c by *negating* that of f^* in c^* :

$$o'(f, c) = -o(f^*, c^*).$$

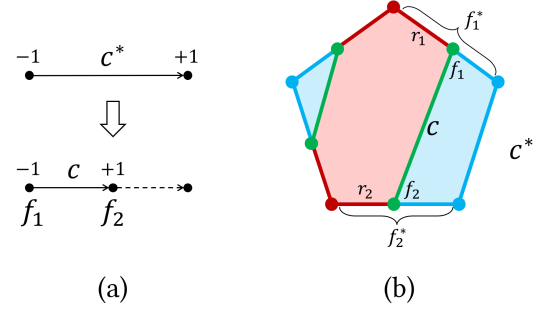


Fig. 5. Illustrations for orientation assignment. Positive, negative and zero-signed cells are colored red, blue, and green, respectively in (b).

We show that the orientation assignment above is consistent.

LEMMA 4.2. *The orientation function o' on the cell complex C' is consistent.*

PROOF. We will verify each criterion (C1-3) in turn. To verify **C1**, consider a 1-cell $c \in C'$ with faces (vertices) f_1, f_2 . It is easy to verify that $o'(f_1, c) = -o'(f_2, c)$ if c is an unsplit edge in C or a 1-room in some edge of C . Otherwise, c is a 1-wall in a 2-cell $c^* \in C$. The orientation $o'(f_i, c)$ ($i = 1, 2$) defined above indicates whether a cyclic traversal on the boundary of c^* transitions from positive h to negative h at f_i . As f_1, f_2 are consecutive on the boundary, they must have opposite orientations.

We next verify criterion **C2** by considering each new k -wall ($k \in [2, n-1]$) or k -room ($k \in [2, n]$) $c \in C'$ (any original cell in C automatically meets the criterion). We need to show that $o'(e, f_1, c) = -o'(e, f_2, c)$ for every subface e of c' shared by faces f_1, f_2 of c . Here, $o'(e, f_i, c)$ is a short-hand for $o'(e, f_i) * o'(f_i, c)$. We consider each type of cell c :

- c is a k -wall in a $(k+1)$ -cell $c^* \in C$ (Figure 4 (a)): Let f_1^*, f_2^* be faces of c^* containing f_1, f_2 . Subface e of c is a wall in the shared face e^* of f_1^* and f_2^* . By our orientation rule, for $i = 1, 2$, $o'(f_i, c) = -o(f_i^*, c^*)$ and

$$o'(e, f_i) = \begin{cases} -o(e^*, f_i^*) * o'(e, r), & k = 2 \\ -o(e^*, f_i^*), & k > 2 \end{cases}$$

where r is the positive 1-room on e^* for $k = 2$. As a result,

$$o'(e, f_i, c) = \begin{cases} o(e^*, f_i^*, c^*) * o'(e, r), & k = 2 \\ o(e^*, f_i^*, c^*), & k > 2 \end{cases}$$

Since $o(e^*, f_1^*, c^*) = -o(e^*, f_2^*, c^*)$ (due to consistency of o), we conclude that $o'(e, f_1, c) = -o'(e, f_2, c)$.

- c is a k -room in a k -cell $c^* \in C$: We separately consider the two cases of the subface e .
 - e is same as a subface e^* of c^* , or is a room in e^* (Figure 4 (b)): Let f_1^*, f_2^* be faces of c^* incident on e^* , which contain f_1, f_2 as rooms. Our construction ensures that $o'(e, f_i, c) = o(e^*, f_i^*, c^*)$ for $i = 1, 2$, and therefore $o'(e, f_1, c) = -o'(e, f_2, c)$ since o is consistent.
 - e is the face of a wall f_1 in c^* (Figure 4 (c)): Let f_2^* be the face of c^* that contains f_2 as a room. By our orientation

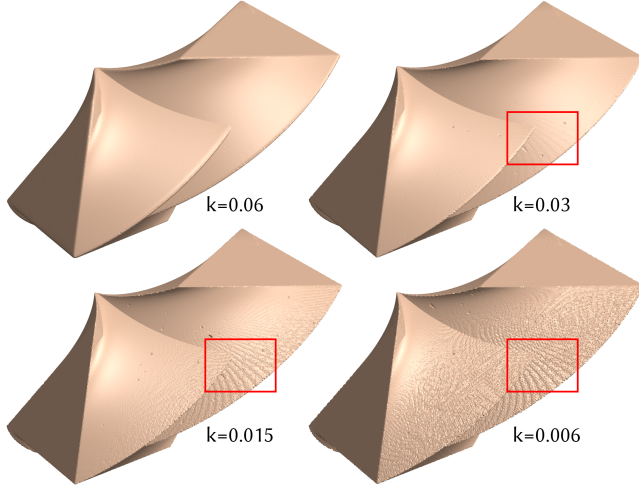


Fig. 6. Sweep surfaces of the rolling tetrahedron example (Figure 2 in paper) computed by [Ju et al. 2025] at different smoothness values k .

rule,

$$o'(e, f_1, c) = \begin{cases} -o(f_2^*, c^*) * o'(e, f_2), & k = 2 \\ -o(f_2^*, c^*) * \text{sign}(h(c)), & k > 2 \end{cases}$$

and

$$o'(e, f_2, c) = \begin{cases} o(f_2^*, c^*) * o'(e, f_1), & k = 2 \\ o(f_2^*, c^*) * \text{sign}(h(c)), & k > 2 \end{cases}$$

Therefore, $o'(e, f_1, c) = -o'(e, f_2, c)$.

Finally, to verify criterion C3, we consider a $(n-1)$ -cell $f \in C'$ that is a face of two n -cells, c_1, c_2 . We need to show that $o'(f, c_1) = -o'(f, c_2)$. Consider two cases:

- f is in some $(n-1)$ -cell $f^* \in C$ (either the same as f^* or a room in it): Each c_i ($i = 1, 2$) is either the same as, or a room in some n -cell $c_i^* \in C$. In both cases, our orientation ensures

that $o'(f, c_i) = o(f^*, c_i^*)$, and hence $o'(f, c_1) = -o'(f, c_2)$ due to the consistency of o .

- f is an $(n-1)$ -wall in an n -cell $c^* \in C$: By Lemma 2.1, c_1, c_2 are n -walls in c^* with opposite h signs. Since $o'(f, c_i)$ is the same as the h sign of c_i , it follows that $o'(f, c_1) = -o'(f, c_2)$. $\square$

5 Soft booleans

The method of Ju et al. [2025] requires the input function to be C^2 -continuous. To test it on CSG sweep functions (which are only C^0), we replace the min/max operations with “soft booleans” defined by cubic polynomials, as described by [Quilez 2013]:

$$\min_{\text{soft}}(a, b, k) = \min(a, b) - \frac{(\max(k - |a - b|, 0))^3}{6k^2}$$

$$\max_{\text{soft}}(a, b, k) = \max(a, b) + \frac{(\max(k - |a - b|, 0))^3}{6k^2},$$

where k is a tunable smoothness parameter that controls the size of the region around the sharp features on the CSG shape that are being “rounded off” (larger k yields more rounding). Both functions have continuous 2nd-order derivatives in a and b .

While a smaller k allows Ju et al. to produce sharper surfaces near feature curves, it also introduces more artifacts and jaggedness on the sweep surface (see highlighted parts in Figure 6). Such artifacts are located on the portion of the envelope projected from the “sharp contours” in 4D, where the gradient of the soft CSG function exhibits a faster transition as k decreases (eventually becoming discontinuous at $k = 0$). As the surfacing algorithm used by Ju et al. relies on smooth interpolation of the values and gradients at the grid points, more abrupt gradient changes lead to larger interpolation errors. In the paper, we used $k = 0.06$ for all our comparisons with [Ju et al. 2025].

References

- Yiwen Ju, Qingnan Zhou, Xingyi Du, Nathan Carr, and Tao Ju. 2025. Lifted Surfacing of Generalized Sweep Volumes. *ACM Transactions on Graphics (TOG)* 44, 6 (2025), 1–17.
- Inigo Quilez. 2013. *Smooth minimum*. <https://iquilezles.org/articles/smin/>